B. Mirror in the String

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: greedy, strings | https://codeforces.com/problemset/problem/1616/B

You have a string $s_1 s_2 \ldots s_n$ and you stand on the left of the string looking right. You want to choose an index k ($1 \le k \le n$) and place a mirror after the k-th letter, so that what you see is $s_1 s_2 \ldots s_k s_k s_{k-1} \ldots s_1$. What is the lexicographically smallest string you can see?

A string a is lexicographically smaller than a string b if and only if one of the following holds:

- a is a prefix of b, but $a \neq b$;
- in the first position where a and b differ, the string a has a letter that appears earlier in the alphabet than the corresponding letter in b.

Input

The first line of input contains one integer t ($1 \le t \le 10\,000$): the number of test cases.

The next t lines contain the description of the test cases, two lines per a test case.

In the first line you are given one integer n ($1 \le n \le 10^5$): the length of the string.

The second line contains the string s consisting of n lowercase English characters.

It is guaranteed that the sum of n over all test cases does not exceed 10^5.

Output

For each test case print the lexicographically smallest string you can see.

Example

Input
4 10 codeforces 9 cbacbacba 3 aaa 4 bbaa

Output
cc cbaabc aa bb

Note

In the first test case choose $k = 1$ to obtain "cc".

In the second test case choose $k = 3$ to obtain "cbaabc".

In the third test case choose $k = 1$ to obtain "aa".

In the fourth test case choose $k = 1$ to obtain "bb".